

GE Healthcare
USA Instruments

GE Signa[®] 1.5T 8-Channel Body Array

Service Manual



2377425-5
Rev 7
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Revision History

Date	Rev	Change Description
10/2003	1	First Issue
11/2005	2	Updated figure 10 to correct landmark position
03/2006	3	Added handle replacement and phantom bottle to element correlation
11/2007	4	Not released - Applied new template. Updated figures 3, 4, 8, 9 and 10. Clarified the correlation between coil element and phantom position at 3-6-1. Cleaned up Contents format and Figures numbering. Added Preamplifier to FRU list. Added notes to symptom #2 of troubleshooting.
05/2008	5	Removed Preamplifier from FRU list, updated symptom #2 of troubleshooting
08/29/08	6	Table 1-1: Updated p/n's. Section 1-3: Revised Op Guide p/n. Section 3-2-3: Updated Step 5. Table 3-2-1, 6-1, & 6-2: deleted USAI p/n column.
9/30/09	7	Updated for the Phantom Elimination Program (DOC0623187).

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Rev. 08/15/2003

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(IT)

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- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

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(LV)

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- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS
(LT)

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ADVARSEL
(NO)

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- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

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(PL)

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- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
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UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT
(TR)

Bu servis kılavuzunun sadece İngilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu İngilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the 8-Channel Body Array, refer to the coil label (see *Figure 1*).

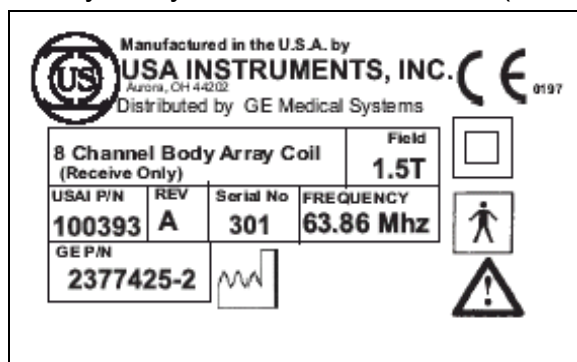


Figure 1: Coil label.

Figure 2 shows a picture of the 8-Channel Body Array.



Figure 2: The 8-Channel Body Array.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	N/A	U1-100393	1
1	Pad Kit (Patient Comfort, Ramp, Filler, Posterior)	2377425-6	U1-150479	1
1	Operator's Guide Packet	2422149-500	N/A	1
1	Service Manual	2377425-5	N/A	1

1-2 Compatibility

This coil is compatible with the Excite 1.5T, Excite HD 1.5T, HDx 1.5T, and HDxt 1.5T systems.

1-3 Related Documentation

Operator's Guide, GE Part Number 2422149.

Signa Excite Service Methods CD, GE Part Number 2333500-3.

1-4 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

Dimensions

Anterior 21.0" L x 27.9" W (53.5cm x 71.1cm)

Posterior 21.0" L x 30.5" W (53.5cm x 77.4cm)

Weight

Anterior 4.4 lb (2.0 kg)

Posterior 8.4 lb (3.8 kg)

1-5 Theory of Operation

The 8-Channel Body Array is designed for high resolution imaging of the human abdomen, both torso and pelvic region. It is a bilateral receive-only 12-element coil, with six elements positioned on both the anterior and posterior coil sections. The block diagrams of the anterior and posterior sections of the coil are shown in *Figures 3 and 4*. *Figure 5* shows the output diagram, the coil to connector interface is shown in *Figure 6* and *Figure 7* shows the block diagram of the circuit layout. The anterior coil is easily detachable from the system cable; the posterior coil is permanently attached to the system cable and system interface connector. The signals from the 12 elements are directed through the use of multiplexing to the 8-channel receiver. All elements are of equal shape and area.

There are three different modes of operation: torso, pelvic and combined. The modes are enabled by the clinical user at the console. In the 8Ch Body Upper mode of operation, the eight upper channels are selected and these are shown in *Figure 8*. Similarly when in 8Ch Body Lower mode the lower eight channels are selected (see *Figure 9*). In the 8Ch Body FullFOV mode all 12 elements are selected (see *Figure 10*). The combined mode is designed for initial scouting and planning purposes. Elements that are not selected during a particular mode are actively decoupled using RF choking networks.

All twelve elements are decoupled during the transmission of the system body coil, with a forward bias DC source provided by the MRI system. Additionally decoupling is also provided by the use of small-signal diodes that are turned on by induced RF voltage in chokes. The active and passive decoupling circuits form high impedance sections in the coil, which segregate the coil circuitry into several isolated electrical segments, preventing any current flow in the coil.

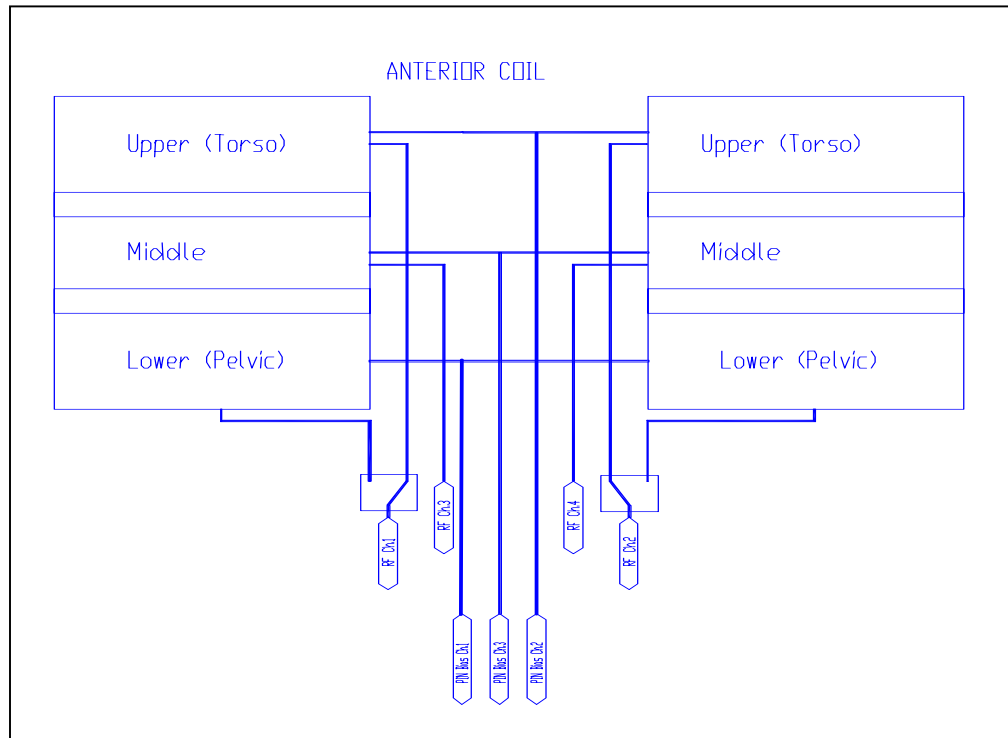


Figure 3: Block diagram of the anterior section of the coil.

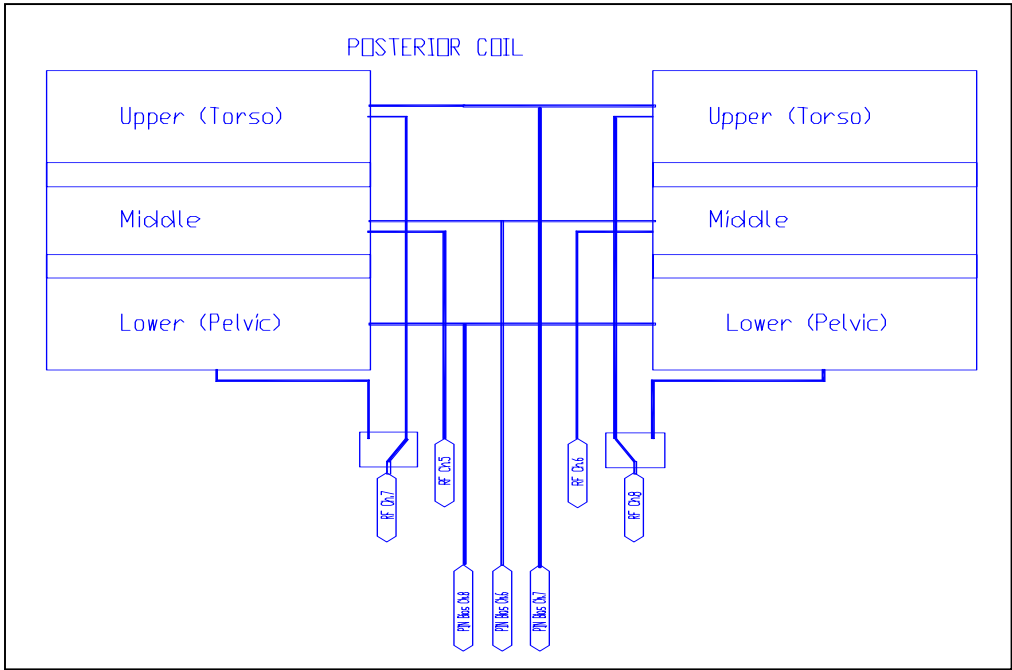


Figure 4: Block diagram of the posterior section of the coil.

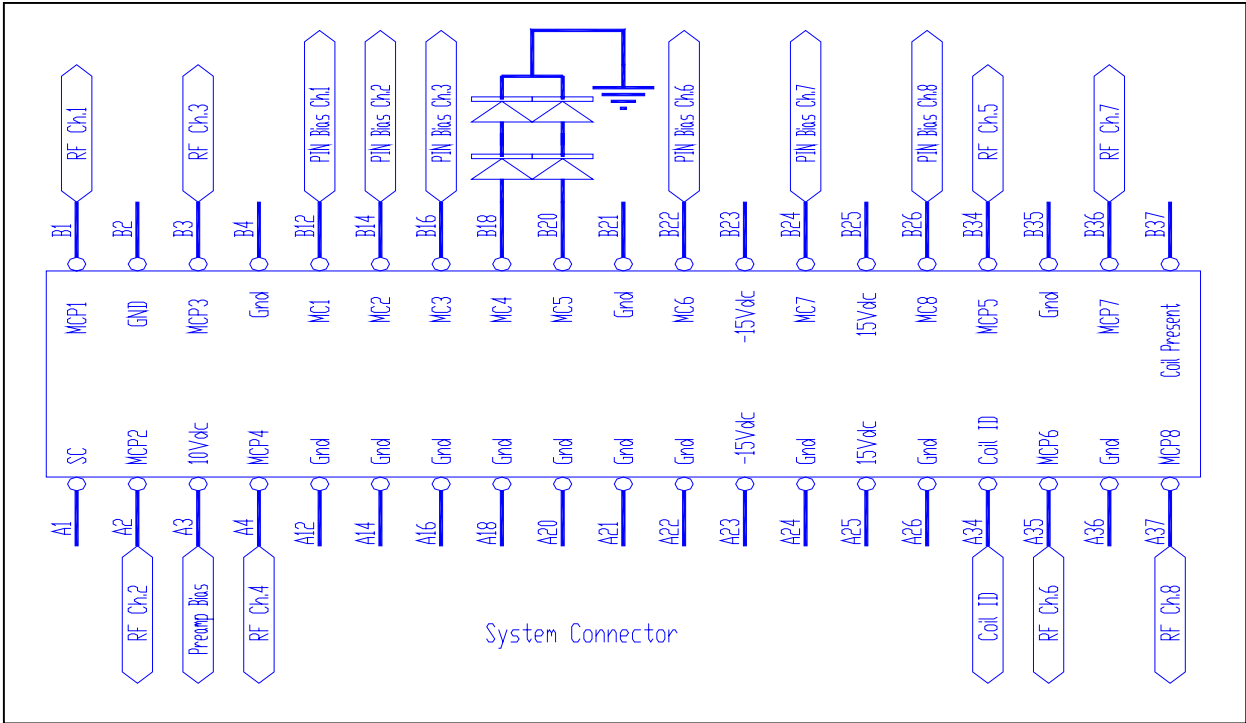


Figure 5: Output diagram.

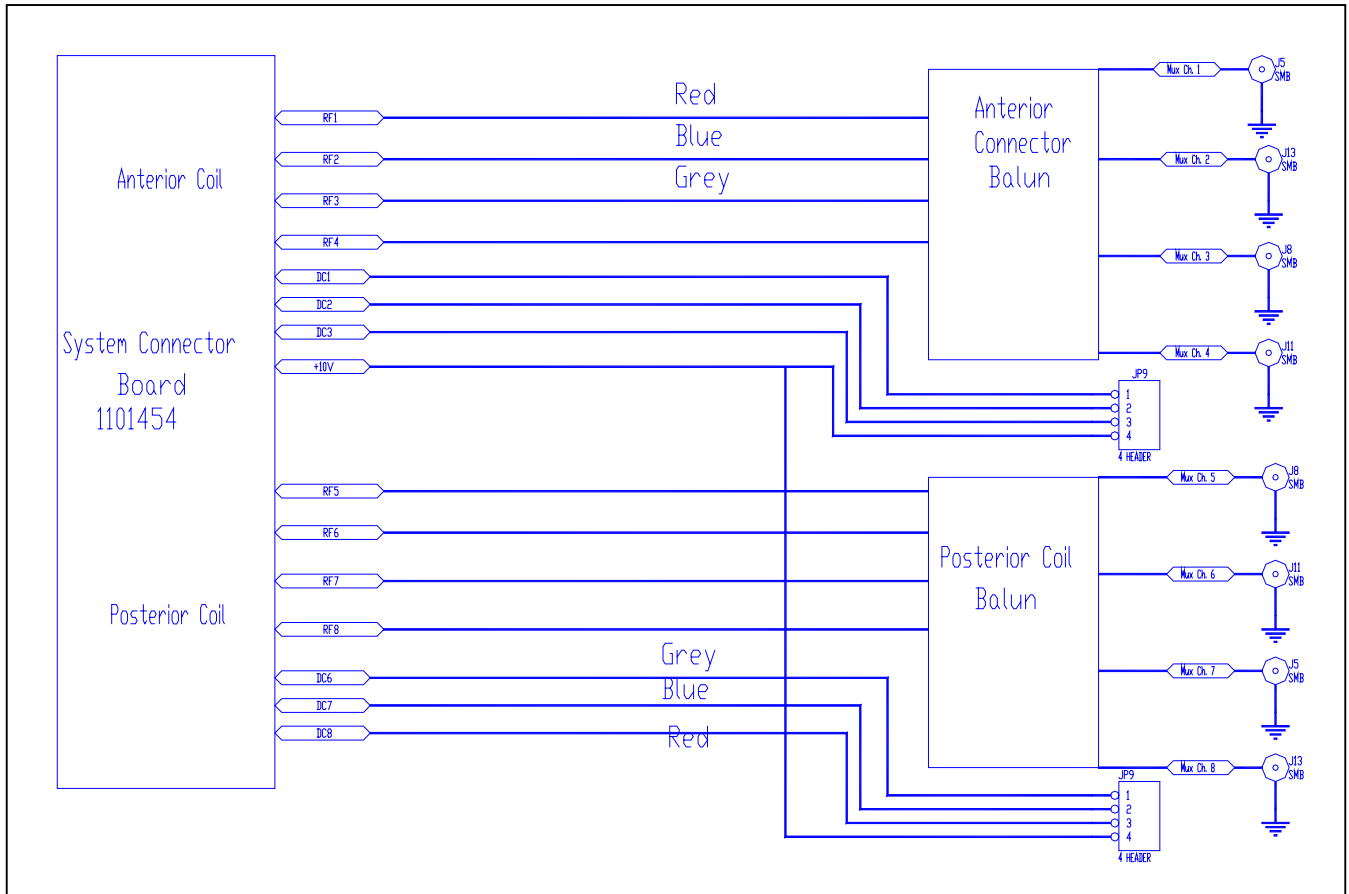


Figure 6: Coil to connector interface.

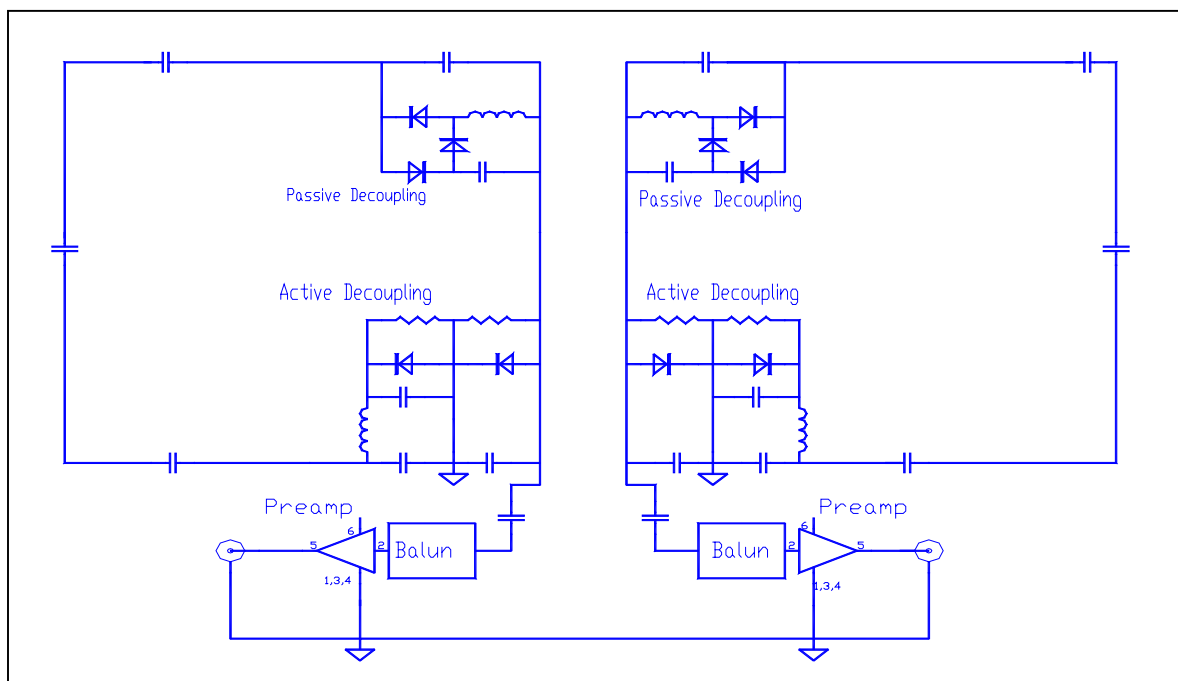


Figure 7: Block diagram of the circuit layout.

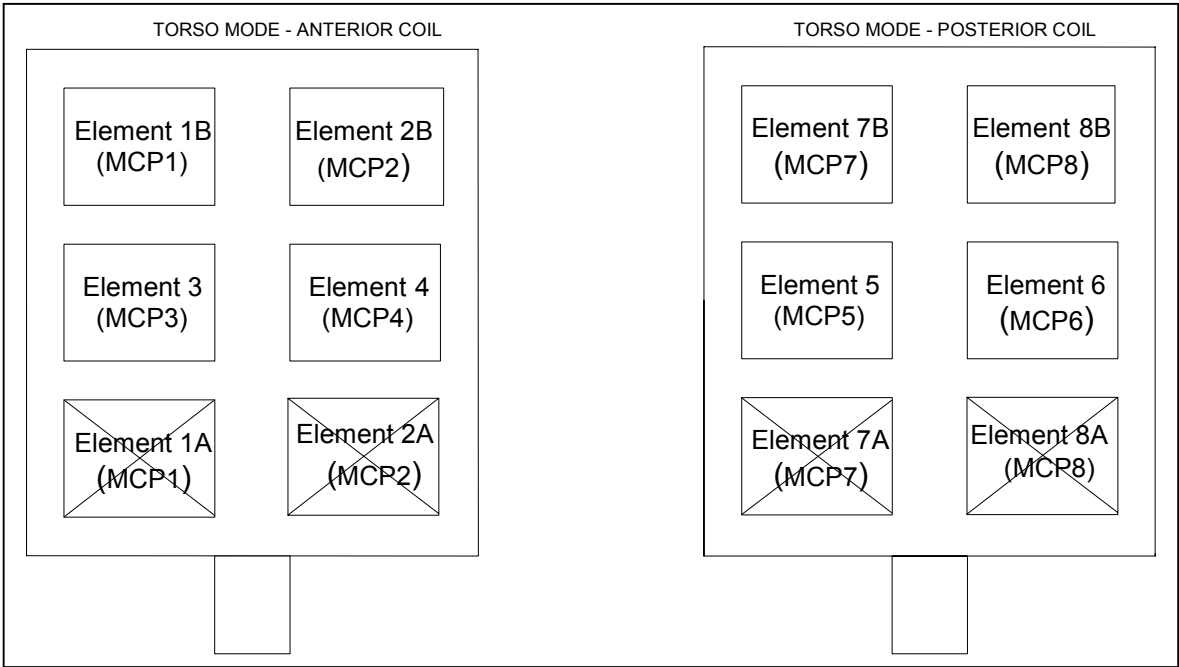


Figure 8: Torso mode of operation (Upper mode).

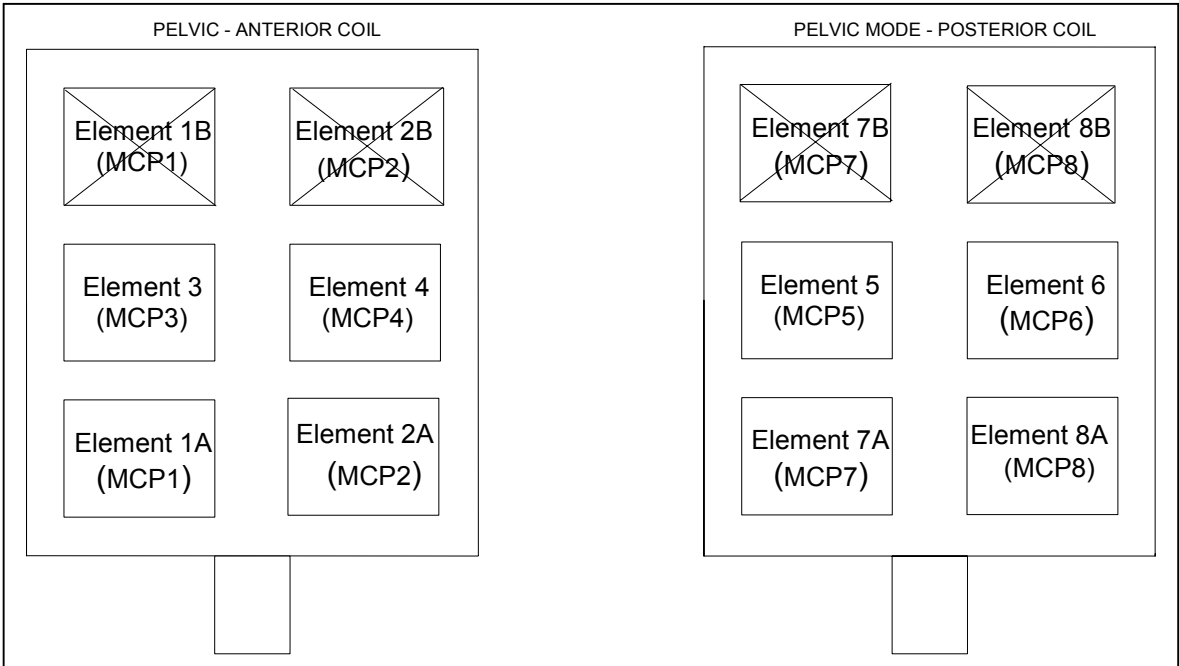


Figure 9: Pelvic mode of operation (Lower mode).

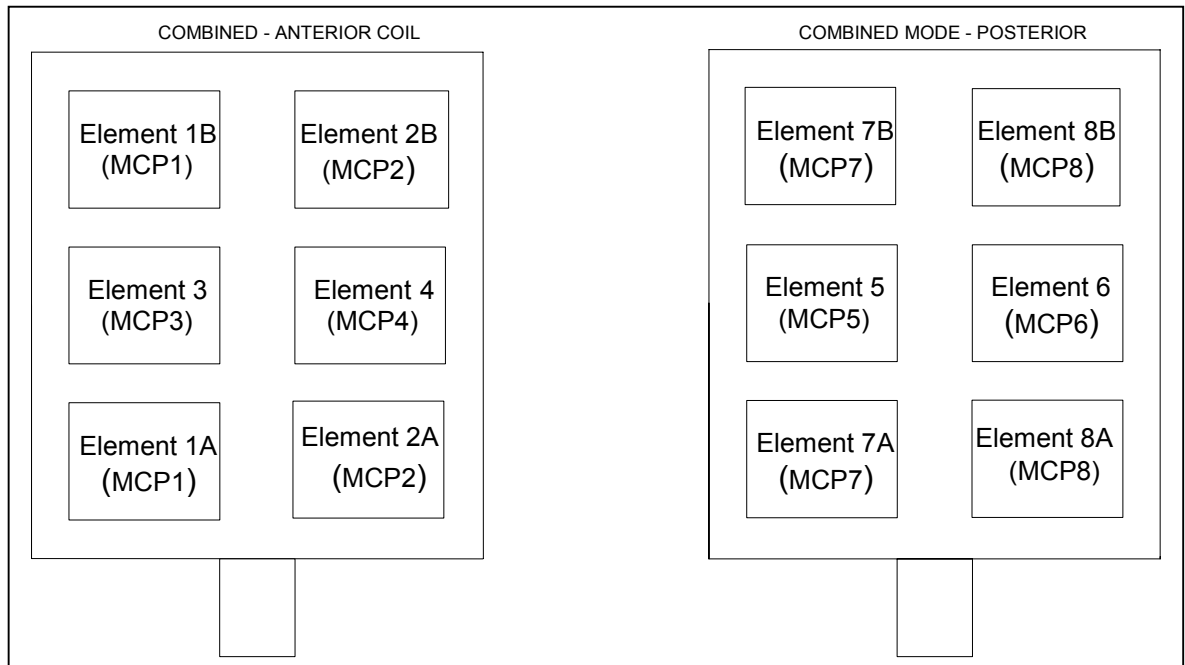


Figure 10: Combined mode of operation (FullFOV mode).

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

None.

2-1-2 Configuration for 11x Software

The system will automatically recognize the coil using the Coil ID feature. No configuration should be required.

2-1-3 Configuration for 10x Software

The 8-Channel Body Array requires 10.0 M4 or greater software. 11.0 M4 software is recommended.

Updated coil configurations in the system are required for software versions prior to 11.0 M4. Updated coil configurations are recommended for 11.0 M4 and greater software revisions.

To update the coil configuration files:

1. Using the configuration manager, remove all currently installed (if any) configurations for the 8-Channel Body Array.
2. Update the coil configurations to the latest version (Refer to Section 9 of “Host Software Install Release 10.X”, direction OW2SCA8G. as found on the Service Methods CD, GE Part Number 2333500-3).
3. Install the new configurations listed in table 2-1-3.

CONFIGURATION MODES – TABLE 2-1-3

8Ch Body Upper	8Ch Body Lower	8Ch Body FullFOV
----------------	----------------	------------------

2-2 Installation Functional Checks

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [**Patient Position**] to open protocols window.
2. Plug the coil into the patient table.
3. At the console, verify the Coil has been properly identified by the system: correct name in **Coil** field and correct picture on the screen.
4. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED USING LEGACY PHANTOM – TABLE 3-2-1

Description	GE Part #	Qty
Phantom Kit	2377425-3	1

TOOLS REQUIRED USING UNIFIED PHANTOM – TABLE 3-2-2

Description	GE Part #	Qty
TL Unified Phantom	5343347	2

3-2-2 Explanation of Procedure

The image quality check uses two consecutive acquisitions for the same slice. The signal is measured at the center of the phantom with a ROI of about 60% of the phantom size. The noise is measured on the subtracted image of the two consecutive signal images using the same ROI. SNR is calculated by dividing the Signal mean by the noise standard deviation. All measurements are performed on composite images. Interim images are acquired for reference only.

3-2-3 SNR Setup and Test Execution for 11.0 M3A and Greater Software Versions with Legacy Phantom

1. Remove any other surface coils (if present) from the cradle.
2. Place the coil on the patient table. Position the phantom set on the coil without the pad, lining up the pegs on the phantom to the holes in the posterior section of the coil (see *Figure 11*).



Figure 11: Positioning the phantom on the coil.

3. Using the raised notches on top of the phantom, align the anterior section on the phantom and fasten the Velcro straps.
4. Landmark the coil at the indicated line and advance to scan (see Figure 12).

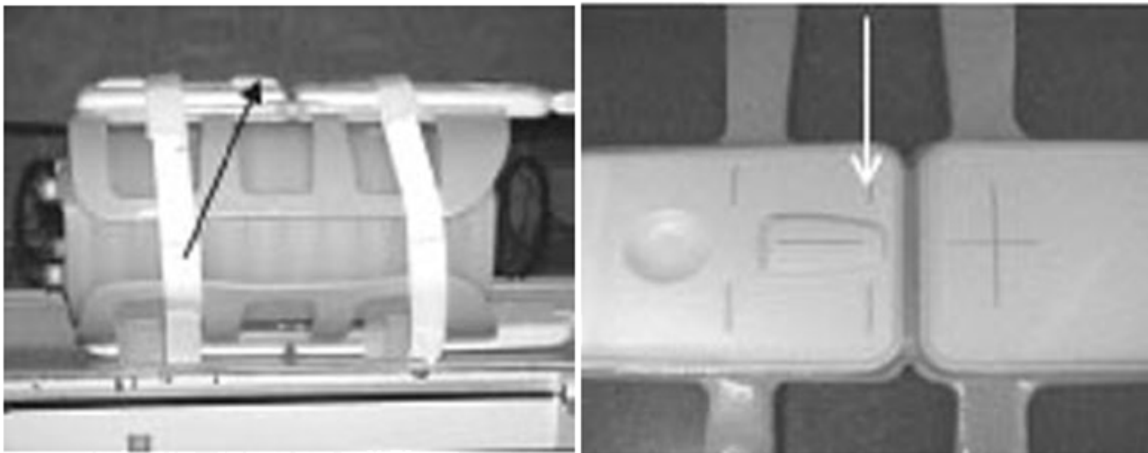


Figure 12: Landmarking the coil.

5. For 12.x or earlier systems: From the Common Service Desktop, select **Image Quality** and then **Coil SNR**. Then **Click here to start this tool**.

For 14.x or later systems: From the Common Service Desktop, select **Image Quality**, and then **MCQA Tool**. Then select **Click here to start this tool**.

6. Select the coil type you want to run, enter the serial number and select **Start Scan**.

Note: Additional documentation can be accessed by selecting **File, Help, [coil type]**.

3-2-4 SNR Setup and Test Execution with Unified Phantom

1. Place the posterior section of the coil on the table.
2. Place the TL unified phantoms over the posterior section as in *Figure 13* such that the TL unified phantoms are well centered on both S/I directions and in the R/L direction.



Top View

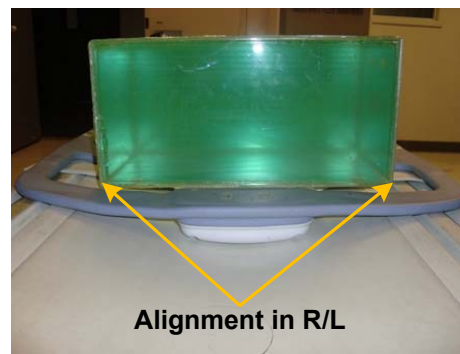
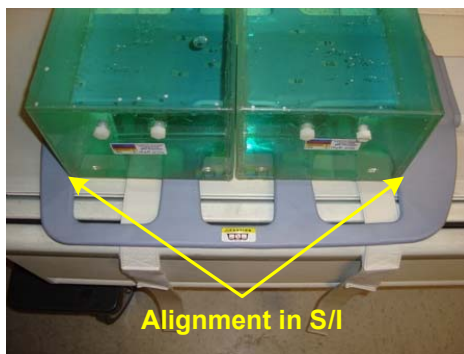


Figure 13: Placing the unified phantoms.

3. Place the anterior section on the TL unified phantoms such that the coil is well centered in both S/I and R/L directions with respect to the phantom set. See *Figure 14*.

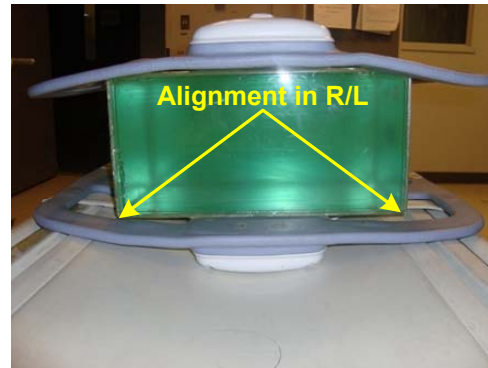
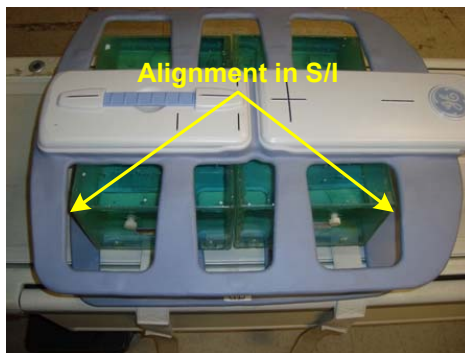


Figure 14: Aligning the anterior section.

4. Fasten the Velcro straps so that the coil firmly snugs the phantoms.

5. Connect the coil to Port A of the system LPCA.
6. Landmark the coil at the center cross mark as shown in *Figure 14* and advance to scan.

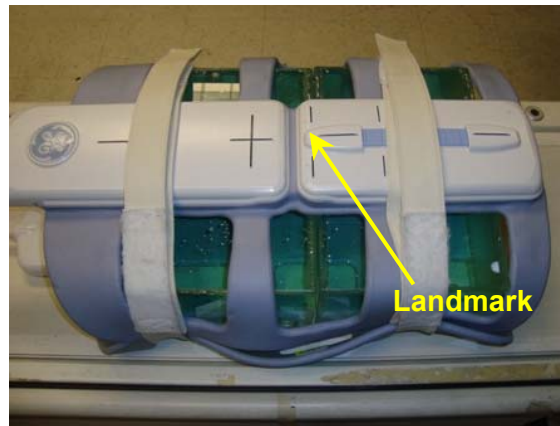


Figure 14: Landmarking the coil.

7. From the Common Service Desktop, select **Image Quality**, and then **MCQA Tool**. Then select **Click here to start this tool**.

3-3 External Cable Check

Diode Check from the Connector:

Step 1 – Using a multi-meter set to the diode check function, check the coil diodes in the following manner.

Step 2 – Orient the connector as shown in *Figure 13*.

Step 3 – Follow Table 3-3 for the correct pins and lead locations to verify the function of the diodes.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3

PIN Number	Positive Lead	Negative Lead	DMM Reading
1	B12	A12	$1.3V \pm 0.2V$
2	B14	A14	$1.3 \pm 0.2V$
3	B16	A16	$0.9 \pm 0.2V$
4	B18	A18	$1.1 \pm 0.2V$
5	B20	A20	$1.1 \pm 0.2V$
6	B22	A22	$0.9 \pm 0.2V$
7	B24	A24	$1.3 \pm 0.2V$
8	B26	A26	$1.3 \pm 0.2V$

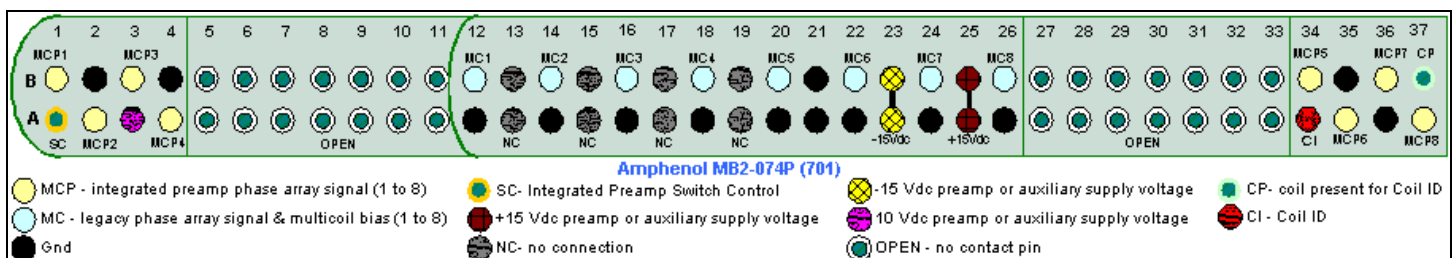


Figure 13: 60 pin Bendix connector.

Note: This connector is system side. Coil side has rows A and B reversed.

3-4 PIN Diodes Check

Same as External Cable Check.

3-5 Troubleshooting Tips

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	If a short or open is observed, check the output cable by itself (see below). If the cable and PIN diodes are okay, then replace the coil.
The output cable has a short or an open.	Disconnect the cable at the coil and check each coaxial line with a DVM (resistance). Try moving and twisting the cable as you watch the meter to find any intermittent connections. Inspect the center pins of the SMB and Bendix connections for wear and proper engagement into the mating connection.	Replace the cable.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs.	Correct MRI system problem.
There is a problem with the coil ID; <i>"The prescribed RF coil and detected RF coil do not match."</i>	Check online instructions by accessing the service browser at Diagnostics; Peripherals; SRI Functional Tests; Coil ID	Start the diagnostic by selecting SRI Coil ID and Run . The diagnostic will display the contents of the coil ID chip whenever the connector is inserted. Follow the troubleshooting instructions if the correct ID is not displayed. When the troubleshooting is complete and the ID is still not displayed, replace the cable.

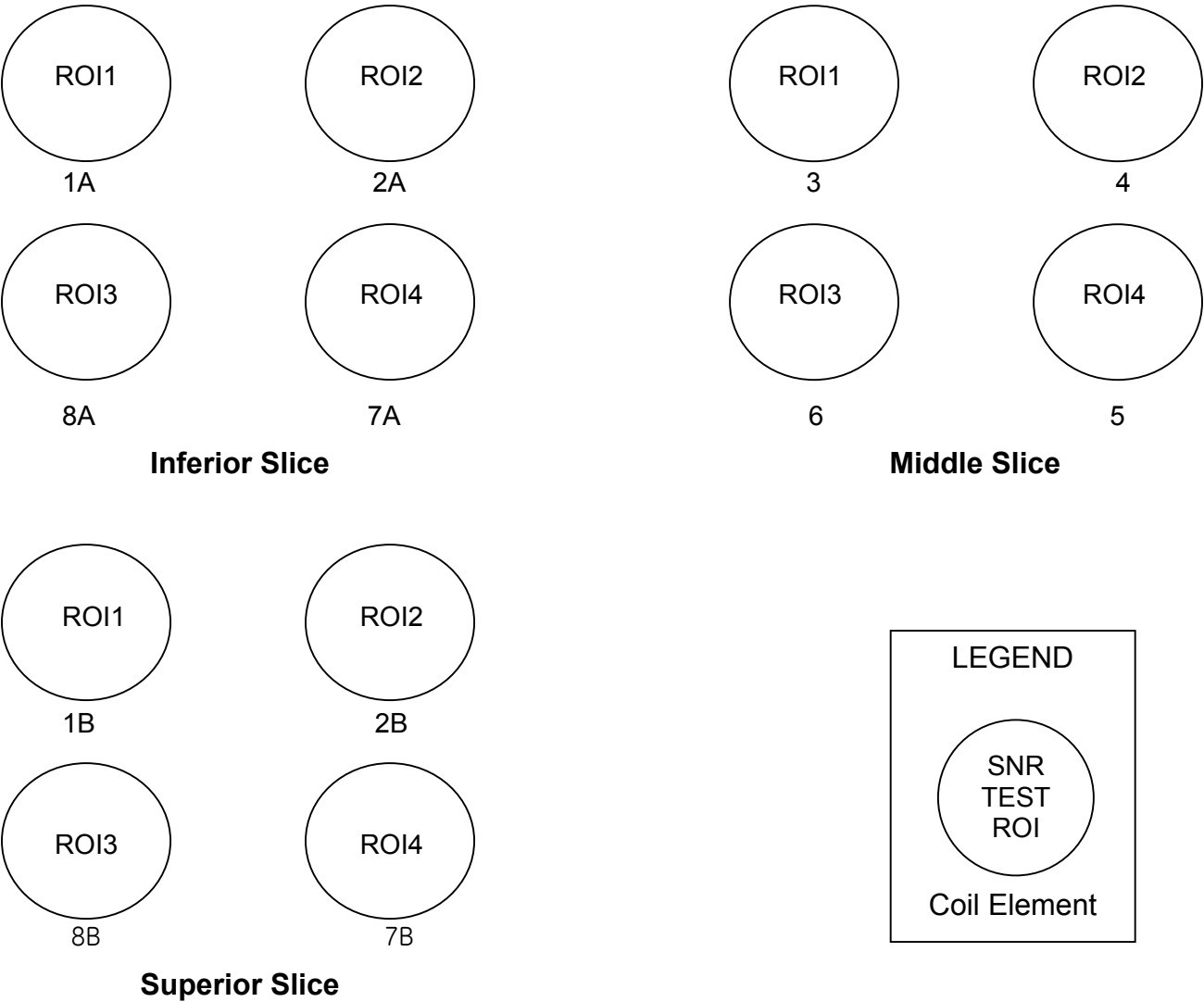
Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise std. dev. measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

3-6 Coil Element Information

Mode	Rcv 1	Rcv 2	Rcv 3	Rcv 4	Rcv 5	Rcv 6	Rcv 7	Rcv 8	PIN 1	PIN 2	PIN 3	PIN 4	PIN 5	PIN 6	PIN 7	PIN 8
8Ch Body Upper	ON	ON	ON	ON	ON	ON	ON	ON	+7	-5	-5	-5	-5	-5	-5	+7
8Ch Body Lower	ON	ON	ON	ON	ON	ON	ON	ON	-5	+7	-5	-5	-5	-5	+7	-5
8Ch Body FullFOV	ON	ON	ON	ON	ON	ON	ON	ON	-5	-5	-5	-5	-5	-5	-5	-5

3-6-1 SNR Phantom position to coil element correlation



SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electrical shock may result if the coil is attached to the system during cleaning or when it is wet.



Do not spray or pour cleaning solution directly on coil. Do not submerge coil in solution. The coil contains sensitive electronics components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. To prevent soiling of the coil, the user should place a cotton sheet over the coil before positioning the patient. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning a coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

Step 1 -- Remove coil cover.

Step 2 -- Remove the two screws that hold down the cable assembly and the two screws that hold down the cable clamps.

Step 3 -- Disconnect SMB plugs and the DC Plug on the cable end from the interface boards.

Step 4 -- Remove the old cable assembly.

Step 5 -- Fix a new cable assembly on the Body Array and reattach the cable lock with two screws. Connect all four SMB plugs and the DC Plug to the corresponding receptacles on the interface board (see *Figure 14 & 15*).

Step 6 -- Close the coil cover.

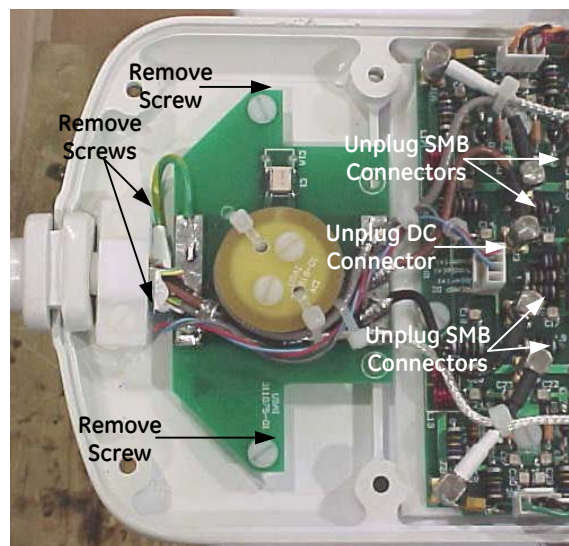


Figure 14: Unfastening the system cable from the posterior coil.

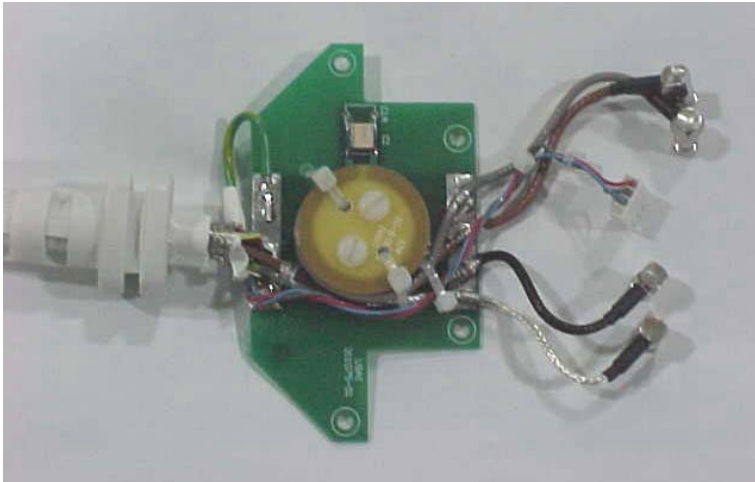


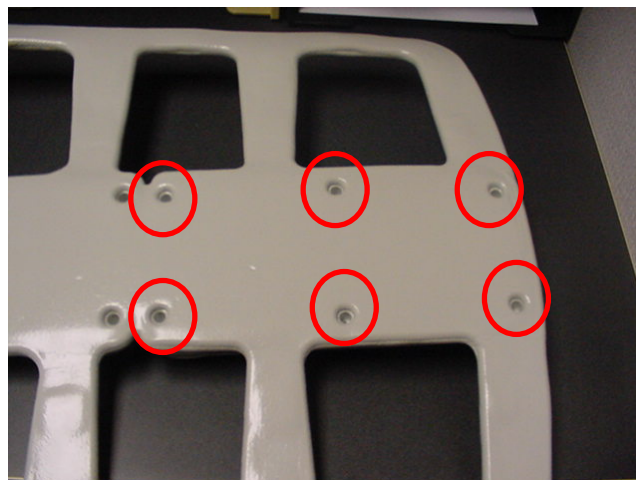
Figure 15: Posterior system cable with balun.

5-2 External Handle Replacement

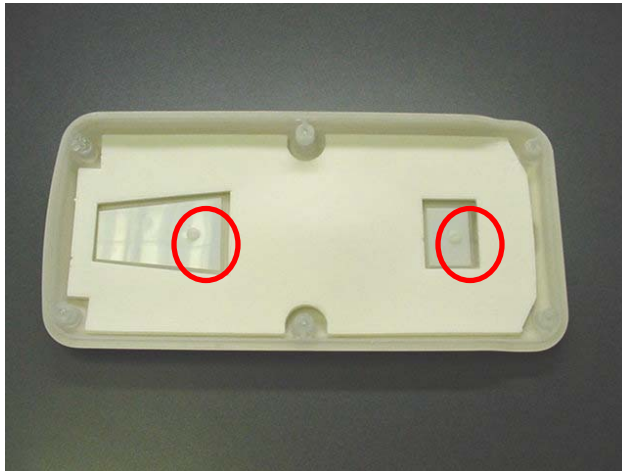
1. Place the anterior section of the coil foam side up on the work area.



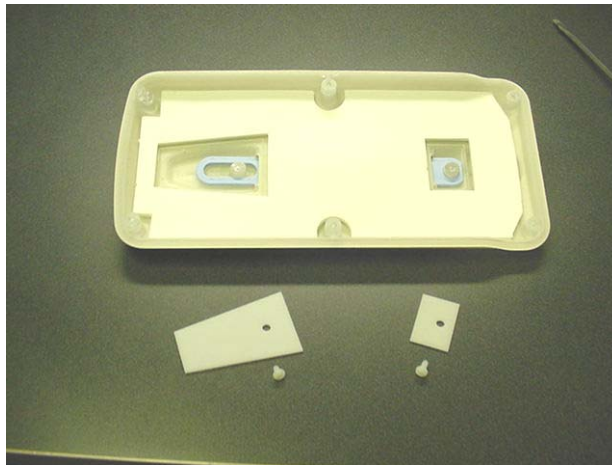
2. Locate and remove the 6 screws holding the smaller anterior cover in place.
Note: Make sure that you do not damage the foam while performing this step.



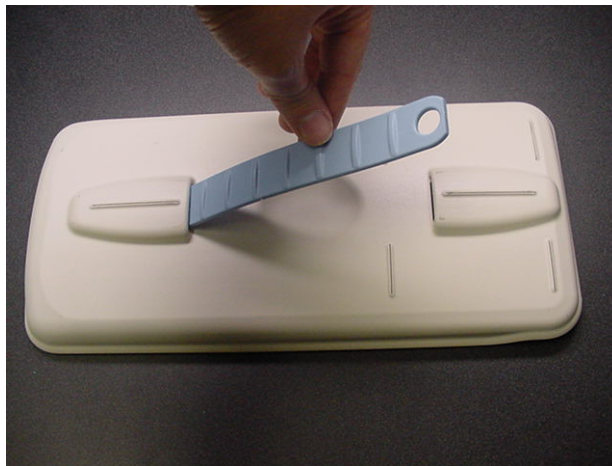
3. Locate and remove the 2 screws holding the ends of the handle on the inside of the cover.



4. Lift the covers to expose the ends of the handle.



5. Lift the ends of the handle off of the posts and pull the handle out of the slots in the cover.
Note: It may be necessary to pry the one end of the handle off of its post.



6. Place the ends of the new handle in the slots of the cover, making sure it is oriented as shown.
7. Place the covers over the ends of the handle and secure them in place with their screws.
8. Place the cover back on the anterior section of the coil and secure it in place by retightening the 6 screws.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #
Coil	2377425-2
Cable Assembly	2377425-7
Phantom Kit (Phantom, Annular Rings, Positioners)	2377425-3
Anterior Coil Handle	2418437
TL Unified Phantom	5343347

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Part Name	GE Part #
Pad Kit (Patient Comfort, Ramp, Filler, Posterior)	2377425-6

SECTION 7 – APPENDIX

7-1 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Date	Configuration	Element	Slice Location	Corresponding Images in Series	Signal Mean	Noise Std. Dev.	SNR	Limit
	8Ch Body Upper							
	8Ch Body Lower							
	8Ch Body FullFOV							

Comments	

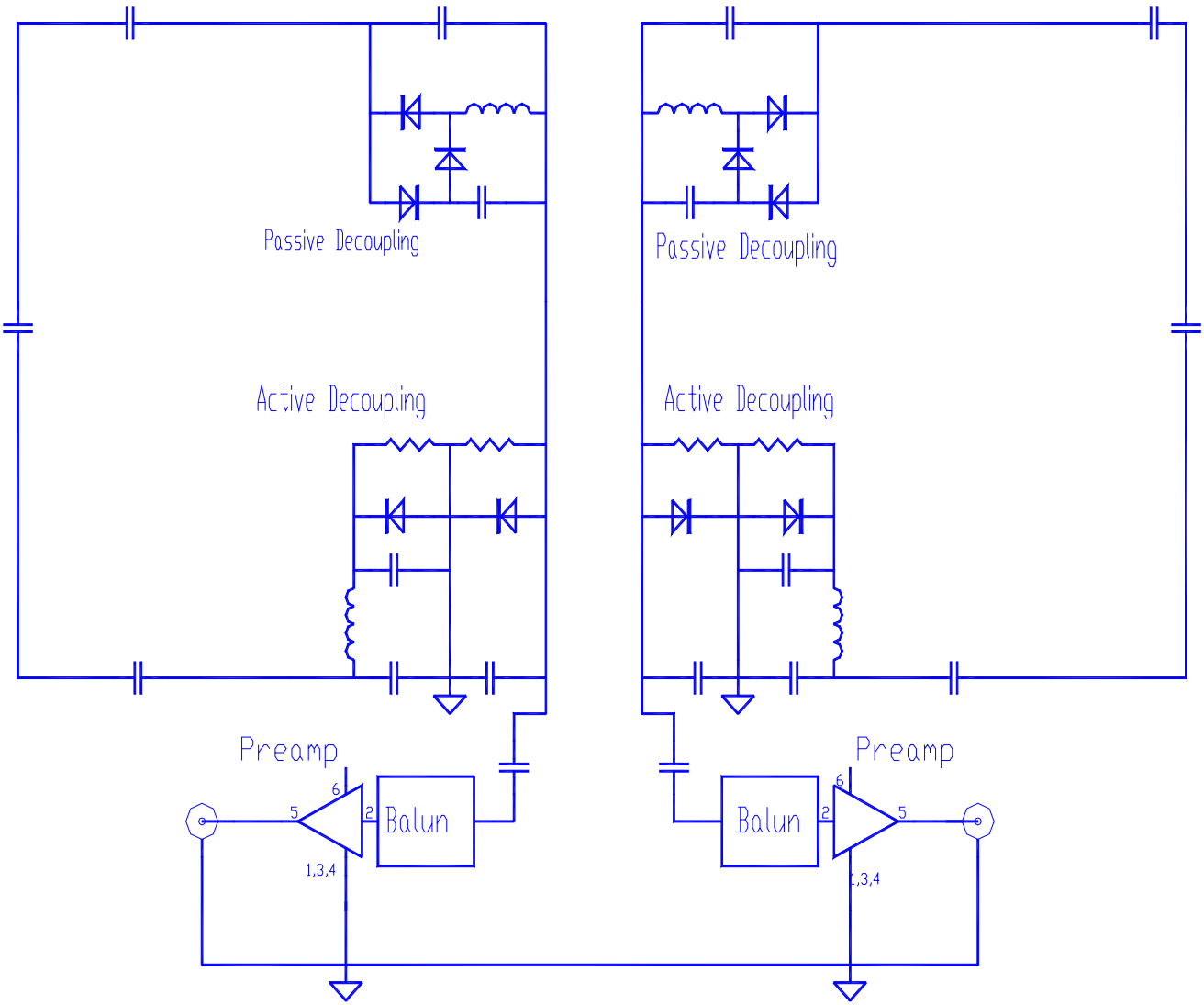
Date	Configuration	Element	Slice Location	Corresponding Images in Series	Signal Mean	Noise Std. Dev.	SNR	Limit
	8Ch Body Upper							
	8Ch Body Lower							
	8Ch Body FullFOV							

Comments	

Date	Configuration	Element	Slice Location	Corresponding Images in Series	Signal Mean	Noise Std. Dev.	SNR	Limit
	8Ch Body Upper							
	8Ch Body Lower							
	8Ch Body FullFOV							

Comments	

7-2 Schematic



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